NETWORKING · NET-07

STP/RSTP & EtherChannel

Use this when: you add redundant links between switches and need to know which port blocks, how fast it recovers, or how to bundle links.

BreakFixLearn

CCNA 200-301 · topics 2.4, 2.5
Sheet v1.0 · Oct 2026

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How STP picks a loop-free tree

Per VLAN with PVST+ / Rapid PVST+

1 Elect the root
Lowest **bridge ID** wins. Tie on priority? Lowest MAC.

2 Root ports
Each other switch: the port with the lowest **cost to root**.

3 Designated ports
Each link: the side closest to root forwards.

4 Block the rest
Leftover ports become **alternate** (or backup) and block.

Bridge ID = | **Priority** 4 bits · default 32768, steps of 4096 | **Extended system ID** 12 bits = VLAN number | **MAC address** 48 bits · tie-breaker

Port cost

Lower = better

SPEED	SHORT (802.1D)	LONG (802.1T)
10 Mbps	100	2,000,000
100 Mbps	19	200,000
1 Gbps	4	20,000
10 Gbps	2	2,000

`show spanning-tree summary` shows the method in use.

STP vs RSTP

Standard	802.1D · **802.1w** (Cisco: Rapid PVST+)
Port states	Blocking, listening, learning, forwarding · **discarding**, learning, forwarding
Timers	Hello 2 s, forward delay 15 s, max age 20 s
Convergence	30 to 50 s · **a few seconds**

Configure STP

```
spanning-tree mode rapid-pvst
spanning-tree vlan 10 root primary
spanning-tree vlan 20 priority 4096
interface g0/5                ! PC or server
 spanning-tree portfast
 spanning-tree bpduguard enable
```

`root primary` sets priority 24576, or lower if needed.
`show spanning-tree vlan 10` shows root, roles and costs.

EtherChannel: bundle links

PROTOCOL	STARTS	WAITS	FORMS?
LACP (IEEE)	active	passive	passive + passive: **no**
PAgP (Cisco)	desirable	auto	auto + auto: **no**
Static	on (no protocol)		only with on

```
interface range g0/1 - 2
 channel-group 1 mode active
interface port-channel 1
 switchport mode trunk
```

Bundle rules

Members	Up to 8 active (LACP: 8 more on standby)
Must match	Speed, duplex, access VLAN or trunk VLANs, native VLAN
Never mix	LACP and PAgP in one bundle
Check	`show etherchannel summary` (want SU / P)

Avoid loops

DANGER PortFast only on end-host ports. On a switch link it can loop the network.

CAUTION Mode `on` skips negotiation; a mismatch can loop.

CAUTION Pick the root on purpose, not by lowest MAC.

REMEMBER

- **Lowest** bridge ID is root
- BPDU Guard err-disables the port
- One bundle = one logical STP link

Sources: IEEE 802.1D, 802.1w, 802.1AX · Cisco Catalyst STP and EtherChannel configuration guides · CCNA 200-301
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